MIL-L-24942

27 SEP 1988

MILITARY SPECIFICATION

LINER, FIREMAN'S TROUSER, QUILTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for removable, quilted heat protective liners for firemen's trousers.

1.2 Classification. The liner covered by this specification shall be of one type only in the following classes, sizes and lengths specified (see 6.2):

1.2.1 Classes.

Class 1 - Regular weight (Aramid batting filler, max. wt. 8.5 oz/yd^2)

Class 2 - Lightweight (Aramid or Novoloid batting filler, max. wt. 4.8 oz/yd^2)

1.2.2 Sizes. 34, 36, 38, 40, 42, 44, 46

1.2.3 Lengths. Regular and Long

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 8415

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, amd handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

V-T-295	- Thread, Nylon
CCC-C-419	- Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered
DDD-L-20	- Label, For Clothing, Equipage, and Tentage (General Use)

MILITARY

MIL-P-15011	- Pallet, Material Handling, Wood, Post Construction, 4 Way Entry
MIL-B-17757	- Box, Shipping, Fiberboard (Modular Sizes)
MIL-T-43636	- Thread, Aramid
MIL-C-43774	- Cloth, Plain or Pajama Check Weave, Aramid
MIL-B-81813	- Batting, Aramid or Novoloid Fiber, Quilted
MIL-B-87002	- Batting, Synthetic Fibers, Quilted, Aramid

STANDARDS

FEDERAL

FED-STD-751	- Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized Unit Loads
MS-27980	- Fastener, Snap, Style 2 (Regular Wire Spring Clamp Type)

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation

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LAWS AND REGULATIONS

U.S. POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superindentent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001).

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, ATTN: Traffic Department, 1616 P Street, N.W., Washington, DC 20036-1404).

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606-5808).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample in which case this specification shall govern.

3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Material.

3.3.1 Basic material.

3.3.1.1 Class 1. The material for the class 1 liner shall be aramid quilted batting conforming to MIL-B-87002 except the use of recycled batting is prohibited. The back outer covering cloth shall be natural in color, conforming to MIL-C-43774.

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3.3.1.2 Class 2. The material for the class 2 liner shall be aramid or novoloid quilted batting conforming to MIL-B-81813 except use of recycled batting is prohibited. The color of the back outer covering cloth shall be natural, and the color for the face outer covering cloth shall approximate Olive Green S-1 or Sage Green. The color of the quilting thread for both needle and bobbin shall approximate Olive Green to match the face outer material.

3.3.2 Thread.

3.3.2.1 Class 1 The thread for the class 1 liner shall be type I, class A, size A of V-T-295. The shade shall approximate that of the dyed outershell.

3.3.2.2 Class 2 The thread for the class 2 liner shall be Olive Green No. S-1, C.A. 66022, sizes E and B conforming to type I of MIL-T-43636. When no standard sample is available the thread shall show "good" fastness to laundering.

3.3.3 Snap fasteners. The snap fasteners for attaching the liner shall consist of a stud and eyelet (male half). The fasteners shall be style 2, 24 line (size 1), nickel plate or bright chrome finish, conforming to MS-27980 and the following part numbers:

Stud - 7N or 7C Eyelet - 8N or 8C

3.3.4 Labels.

3.3.4.1 Identification/ instruction label. Each liner shall have an identification/ instruction label conforming to type VI, class 15 of DDD-L-20, except that the colorfastness requirements shall not apply. The label shall bear the following inscription:

Class 1

LINER, FIREMAN'S TROUSER, QUILTED (Class 1)
BASE FABRIC: RAYON FACE
ARAMID BATTING
ARAMID BACK
CONTRACT NO. DLA-100-00-0-0000 (EXAMPLE)
NAME OF CONTRACTOR:
NAME OF MANUFACTURER: (If other than contractor)

CLEANING INSTRUCTIONS

Machine Wash Cold with Detergent
Tumble Dry - LOW HEAT

OR
Follow NAVEDTRA 414-01-45-81, Formula III
USE COLD WATER WASH AND LOW TUMBLE DRY TEMPERATURE

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Class 2

LINER, FIREMAN'S TROUSER, QUILTED (Class 2)
BASE FABRIC: 100% ARAMID
CONTRACT NO. DLA-100-00-0-0000 (EXAMPLE)
NAME OF CONTRACTOR:
NAME OF MANUFACTURER: (If other than contractor)

CLEANING INSTRUCTIONS

Hand wash or machine wash cold.

Tumble dry - low heat.

3.3.4.2 Size and stock number label. Each liner shall have a size and stock number label conforming to type VI, class 2 of DDD-L-20, except that the colorfastness requirements shall not apply. The label shall bear the following inscription:

SIZE: 38 R (EXAMPLE)
STOCK NUMBER: 8415-00-000-0000 (EXAMPLE)

NOTE: The lengths (Regular and Long) may be abbreviated as R or L.

3.3.4.3 Warning label. Each liner shall have a warning label conforming to type VI, class 9 of DDD-L-20 stitched to the inside back of the liner. The label shall be printed with a red marking medium. The label shall be printed with block, sansserif type with letters 1/2 inch in height. The label shall contain the following inscription:

WARNING
MUST BE WORN WITH THIS SIDE TOWARDS BODY

Two lines may be used for the warning inscription.

3.3.5 Facing. The waist facing for the liner shall be fabricated from an unbleached cotton duck, size number 6, medium texture conforming to type I of CCC-C-419.

3.3.6 Binding.

3.3.6.1 Class 1 The binding for the raw edges of the class 1 liner shall be a commercial grade bias-cut rayon binding with a minimum width of 2-5/8 inches.

3.3.6.2 Class 2 The binding for the raw edges of the class 2 liners shall be a bias cut, 3 oz/sq yd aramid cloth conforming to MIL-C-43774, and have a minimum width of 2-5/8 inches.

3.4 Design. The liner shall have snap fasteners along the edges of the waist and leg openings. The liner shall be made of quilted material which affords thermal protection. The edges of the liner are bound (see Figure 1).

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3.4.1 Figure. The figure is furnished for information purposes only. When inconsistencies exist between the written specification and the figure, the written specification shall govern.

3.5 Patterns. Standard patterns to be used to cut working patterns will be furnished by the Government (see 6.3). The patterns allow a 3/8 to 1/2 inch seam allowance for joining seams. The working patterns shall be identical to the Government patterns. Neither the Government patterns nor the working patterns shall be altered in any way except that additional notches for use during construction are allowed on the working patterns.

3.5.1 Pattern parts. The component parts of the liner shall be cut from the materials specified in accordance with the following pattern parts:

Material	Pattern parts	Cut parts
Quilted batting	Back	2
	Front	2
Cloth, duck, cotton	Waist facing	2

3.6 Construction.

3.6.1 Stitch, seam, and stitching types. Stitch, seam, and stitching types specified in Table I shall conform to FED-STD-751. Wherever two or more methods for seam or stitch types are given for the same part of the operation, any one may be used. Where stitch type 401 is used, the ends of the stitching shall be caught in other seams or row(s) of stitching and the looper (underthread) shall be on the outside. The bight of the overedge stitching shall be 3/16 to 1/4 inch. A knife shall be used on the overedging machine with the guide set so that the edges of fabric are not cut off, but only the ravelled edges are trimmed.

3.6.2 Thread breaks and ends of seams. Ends of all seams and stitchings when not caught in other seams or stitchings shall be backstitched or overstitched not less than 3/8 inch. The ends of a continuous line of stitching shall be overlapped not less than 1/2 inch. Thread breaks or skipped stitches (301 or 401 stitch types) may be repaired by 301 stitch type with the stitching anchoring both ends of the break or skipped stitches not less than 1/2 inch.

3.6.3 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Table I.

3.7 Manufacturing operations requirements. The liner shall be manufactured in accordance with the operation requirements specified in Table I. The contractor is not required to follow the exact sequence of operations listed provided that the finished liners are identical to those produced by following the sequence of operations as listed in Table I.

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3.7.1 Size marking. The component parts of the liner shall be bundled, marked, or ticketed to assure proper size assembly. The use of metal fastening devices or sewn-on tickets for marking is prohibited. Adhesive tickets, when used, shall not discolor the material and the adhesive mass shall not adhere to the material upon removal of the tickets.

NOTE: The use of ink pad numbering machine, rubber stamp, or pencil is allowed.

3.7.2 Abbreviations in table of operations. The abbreviations used in Table I are as follows:

Stch	-	Stitch
in	-	inch
Ndl	-	Needle
Bob	-	Bobbin
Lpr	-	Looper
Mchne	-	Machine
Brtck	-	Bartack
Comrcl	-	Commercial
smlr	-	similar
Btnhl	-	Buttonhole
incl	-	including
dbl	-	double

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TABLE I - CONSTRUCTION OF LINER

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL BOB/ LPR
	NOTE: The dyed outer covering of the batting is considered the face, and is worn against the wearer when snapped into position inside the appropriate trouser.				
1.	Cutting a. The liner shall be cut in strict accordance with patterns furnished which show size, placement of component parts, directional lines and notches for proper assemble. A maximum swing of 2-1/2 inches from the directional line is permitted. Measurements shall be taken at the top and bottom of the directional line on the pattern to the selvage edge of the fabric on the even side of the lay. The difference between the two measurements shall not exceed 2-1/2 inches. Drill holes for positioning are permitted provided they are covered in the finished liner. b. The component parts of the quilted liner may be cut from ends, and shall approximate each other in shade within a garment.				
2.	Replacement of damaged parts During the spreading, cutting, and manufacturing operations, components having material defects or damages that are classified as defects in section 4 shall be removed from production and replaced with nondefective and properly matched parts.				
3.	Size marking (See 3.7.1) Mark, bundle, or ticket the component parts to assure proper size assembly.				
4.	Join liner. a. Join the outseams with notches matching, face to face.	301 or 401	SSa-1	6-8	

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TABLE I - CONSTRUCTION OF LINER

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL BOB/ LPR
	b. Overedge stitch the raw edges together.	502 or 503 or 602	SSa-1	6-8	
	c. Join the seat seam, face to face.	301 or 401	SSa-1	6-8	
	d. Overedge stitch the raw edges together.	502 or 503 or 602	SSa-1	6-8	
	e. Join the crotch seam, face to face.	301 or 401	SSa-1	6-8	
	f. Overedge stitch the raw edges together.	502 or 503 or 602	SSa-1	6-8	
	g. Join the inseams in one operation, face to face, catching the seat and crotch seams in the stitching.	301 or 401	SSa-1	6-8	
	h. Overedge stitch the raw edges together.	502 or 503 or 602	SSa-1	6-8	
	OR				
	As an alternate, operations 4a and b, 4c and d, 4e and f, and 4g and h, may be performed in one operation by simultaneously stitching and overedging.	515 or 516 or 519	SSa-2	6-8	
5.	Attach waist facing				
	a. Stitch waist facings at center back 3/8 inch from raw edges, face to face.	301	SSa-1	6-8	
	b. Stitch 3/16 inch from raw edge, the back side of waist facing piece and face side of the removable liner at top of quilted liner.	301	SSa-1	6-8	
	c. Turn under 3/8 inch the lower raw edge of waist facing and topstitch through all plies.	301	LSd-1	6-8	

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TABLE I - CONSTRUCTION OF LINER

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL BOB/ LPR
6.	Bind liner Bind the edges of waist, fly opening and leg bottoms of the liner with appropriate binding as specified in 3.3.6. All raw edges of the binding shall be turned under and the finished dimension from the outside folded edge to turned edge shall be 1 (+1/8,-0) inch. Stitching shall be 7/8 (+1/16,-0) inch from folded edge. On leg bottoms overlap the ends of binding 3/4 to 1-1/4 inches. The raw edges of the lining assembly shall extend to the deepest fold of the binding.	301	BSa-1	6-8	
7.	Attach labels a. Center the identification/instruction label on the seat seam on the inside back (against wearer) with the top edge of label 1 to 1-1/2 inches below the waist facing. Stitch all four sides of label 1/16-1/8 inch from edge.	301	LSbj-1	6-8	
	b. Center the size and stock number label directly below the identification/ instruction label and stitch all four sides of label 1/16 to 1/8 inch from edge. Lower edge stitching of identification/ instruction label may be used to secure the top edge of size and stock number label.	301	LSbj-1	6-8	
	c. As an alternate, the two labels may be combined, provided all inscriptions are maintained.				
	d. Center warning label approximately 1-1/2 inches below size and stock number label and stitch all four sides of label 1/16 to 1/8 inch from edge.	301	LSbj-1	6-8	

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TABLE I - CONSTRUCTION OF LINER

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL BOB/ LPR
8.	Attach snap fasteners a. Position and attach nine stud snap fasteners, as indicated by marks on patterns, to the waistband. Studs shall face the inside of the liner (towards the wearer). b. Position and attach two stud snap fasteners to each leg bottom, one on inseam and one on outseam. Studs shall face the outside of liner. Center of stud shall be 1/2 to 5/8 inches from bound edge.				
9.	Clean liner a. Trim all thread ends. b. Remove all loose threads. c. Remove all spots and stains.				

NOTE: Care shall be taken when trimming thread ends,so that damage to the fabric does not result.

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3.8 Sizes and measurements. Sizes and measurements of the finished liners shall be as shown in Table II (see Figure 1). All measurements and tolerances are expressed in inches.

Table II - Finished Measurements

SIZE	(A) Waist	(B) Seat	(C) Inseam Regular	(C) Inseam Long
34	34	46	17	22
36	36	48	18	23
38	38	50	18	23
40	40	52	18	23
42	42	54	18	23
44	44	56	18	23
46	46	58	19	24
Tolerance	±1	±1	± 1/2	±1/2

NOTE: The liner shall be laid flat and smooth with the top edges of fly opening abutting.

(A) Waist - Twice the measurement taken across front 1-1/4 inches down from top of waistband from stitch line to stitch line with front edges abutted.
(B) Seat - Twice the measurement taken straight across the front from inside stitch line to inside stitch line 1 inch below bottom of fly.
(C) Inseam - The measurement shall be taken along the inseam from the junction of the front center seam and the seat seam to the bottom of the liner legs.

3.9 Workmanship. The finished liners shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required, the first article submitted in accordance with 3.2, shall be inspected as specified in 4.4.2 for compliance with design, construction, workmanship, appearance and dimensional requirements.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents.

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with 4.4.3 through 4.4.4.

4.4.3 End item visual examination. The end item shall be examined for the defects listed in Table III. The sample unit shall be one finished trouser liner. The lot size shall be expressed in units of trouser liners. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects, 15.0 for Major and Minor A defects, and 40.0 for total (major, minor A&B combined) defects.

NOTE: The "inside" of the liner is the dyed outer covering of the batting which will face the wearer when snapped into position of the appropriate trouser.

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Table III. End item visual defects

Examine	Defect	Classification Major	Minor A	Minor B
I. General	a. Any hole, cut, tear, needle chew, smash, multiple float (In any ply)	X		
	b. Darn or mend (In any ply)		X	
	c. Exposed drill hole (In any ply)			X
	d. One or more rows of stitching omitted in the quilting	X		
	e. More than 1-1/2 inches of broken or missing stitches in quilting			X
	f. Thick or thin place in batting	X		
	g. Piecing or splicing of either ply of quilting	X		
	h. Any part not cut in accordance with directional lines indicated on patterns or not in accordance with specifications	X		
	i. Three or more thread ends 1/4 inch or longer not trimmed on inside of liner		X	
	j. Five or more thread ends 1/4 inch or longer not trimmed on outside of liner			X
	k. Spot or stain on inside of liner:			
	1. Up to 1 inch in longest direction			X
	2. One inch or more in longest direction		X	
II. Components and Assembly	a. Any component not as specified or any defect of assembly	X		
	b. Any component, component part, or required operation omitted, or any operation improperly performed.	X		
	c. Aramid cloth not finished on outside of liner		X	
	d. Color of component not as specified	X		
III. Seams and Stitching	a. Accuracy of seaming:			
	1. Any seam, hem or attachment of any part puckered, pleated, or caught in an unrelated operation or stitching		X	
	2. Any seam allowance not as specified			X
	3. Any end of seam and stitching when not caught in another seam or stitching:			
	(a) Not backstitched		X	
	(b) Backstitched or overstitched less than 3/8 inch			X
	4. Any seam restitched damaging cloth to a degree where it may tear along the row of stitching	X		
	5. End of continuous line of stitching overlapped less than 1/2 inch			X
	6. Stitching anchoring both ends of thread break or skipped stitches less than 1/2 inch			X

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Table III. End item visual defects

Examine	Defect	Classification Major	Minor A	Minor B
III. Seams and Stitching cont.	b. Open seam:			
	1. Any seam except quilt or serge stitching		X	
	2. Quilt or serge stitching - 1 inch or more			X
	Note: A seam shall be classified as open when one or more stitches are broken or two or more continuous runoff or skipped stitches occur.			
	c. Seam and stitch type:			
	1. Not specified seam or stitch type			X
	2. Looper (underthread) on outside when 401 stitch type is used		X	
	d. Stitches per inch:			
	1. Stitches more or less than specified:			
	(a) By more than two stitches	X		
	(b) By more than one stitch			X
	Note: Score only when condition exists on more than 1/2 the length of the seam			
	e. Overedge stitching:			
	1. Omitted for 2 or more inches on one or more seams			X
	2. Bight of stitching less than 3/16 or more than 1/4 inch			X
	f. Bartack or backtack:			
	1. Missing, insecure, misplaced or not serving intended purpose		X	
	2. Loose stitch tension, not specified length or type			X
	g. Stitch tension:			
	1. Loose tension resulting in loosely exposed loops of lower or top thread			X
	2. Tight tension (stitches break when normal strain is applied in the direction of the seam or stitching):			
	(a) Quilt or serge stitching			X
	(b) Liner stitching		X	
	h. Stitch margin:			
	1. Not within specified range			X
	2. Top or edge stitching sewn close to edge resulting in damage to the material:			
	(a) Affecting appearance			X
	(b) Affecting serviceability	X		

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Table III. End item visual defects

Examine	Defect	Classification Major	Minor A	Minor B
IV. Snap Fasteners	a. Not specified type, size or finish	X		
	b. Studs and sockets not positioned or aligned as indicated on pattern		X	
	c. Missing, broken or defective	X		
	d. Not properly set or attached:			
	1. Will adequately be retained on garment and will function as intended			X
	2. Will likely become detached from garment in a short time of use		X	
	Note: Snaps shall be checked for proper function and secure clinching by opening and closing several times.			
V. Binding	a. Binding not securely caught, pulls away from quilted liner		X	
	b. Binding noticeably puckered or pleated		X	
	c. Binding tight, will not lie flat		X	
	d. Binding ends overlapped other than specified			X
	e. Width of binding other than specified			X
	f. Binding not as specified		X	
VI. Label	a. Missing, incomplete, incorrect, or illegible		X	
	b. Insecure, i.e., not stitched on all four sides			X
	c. Position not as specified			X

4.4.4. Dimensional examination of the trouser liner. The trouser liners shall be examined for dimensional defects. Any dimension not meeting the specified requirement shall be classified as a defect. The lot size shall be expressed in units of trouser liners. The sample unit shall be one trouser liner. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0. Width and length measurements shall be made while the trouser liners are placed on a flat surface with wrinkles and creases smoothed out. No tension shall be applied when making width and length measurements. Measurements shall be made as specified in Table II.

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4.4.5 Examination of packaging requirements. An examination shall be made to determine that packaging, packing and marking comply with Section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery. Defects of closure listed below shall be examined on shipping containers in the end item inspection lot. The inspection level shall be S-2 and AQL shall be 2.5 defects per hundred units.

Examine	Defect
Marking (exterior and interior	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, improper taping, inadequate stapling, bulged or distorted containers.
Content	Number of items per shipping container is more or less than required. Size shown on one or more items not as specified on shipping container. 1/

1/ For this defect, one item from each shipping container in the sample shall be examined.

4.4.6 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5 as specified in MIL-STD-105.

Examine	Defect
Finished dimension requirements.	Length, width, or height exceeds specified maximum
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

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5.1.1 Level A. Each liner shall be folded as follows:

1. Fold liners in half lengthwise with the fly on the inside.
2. Fold liners in half again by bringing the bottom of liner up to the top.

The folded liner shall measure approximately 15 inches by 15 inches. Each folded liner shall be individually packaged in a snug-fitting, clear polyethylene film bag having a film thickness of not less than 0.00125 inch. The bag may be fabricated from polyethylene film tubing or sheeting. All seams and closures shall be formed by heat sealing. A 1/4 inch diameter hole shall be made at one corner of each bag to allow excess air to escape. As an alternate, the polyethylene bag may be of the tuck-in or reverse flap type, in which a heat seal closure and corner hole are not required.

5.1.2 Level C. Each liner shall be packaged to afford adequate protection against physical damage during shipment from the contractor to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Ten liners, of one class, size, and length only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed, and reinforced conforming to type CF, class weather-resistant, grade V15c, variety DW, size 3A of MIL-B-17757. The fiberboard used for the liner shall conform to type CF, class domestic conforming to MIL-B-17757. Level A folded liners shall be stacked on edge within the shipping container with the face of the packages parallel to the end panels within the shipping container.

5.2.2 Level B. Ten liners, of one class, size, and length only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed and reinforced conforming to type CF, class domestic, variety DW, grade 275, size 3A of MIL-B-17757. The fiberboard used for the liner shall conform to type CF, class domestic conforming to MIL-B-17757. Level A folded liners shall be stacked on edge within the shipping container with the face of the packages parallel to the end panels within the shipping container.

5.2.3 Level C. Liners packaged as specified in 5.1 shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with U.S. Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

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5.3 Palletization. When specified (see 6.2), item, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be banded with primary and secondary straps in accordance with the banding means C, K and L, or O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. The pallet shall be 4-way, type IV; type V, class 1, size 2; or type VIII, fabricated from wood groups I, II, III or IV, grade A of NN-P-71, or 4-way, style 1, size A, type I, class 1 fabricated from groups specified of MIL-P-15011. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contacting officer.

5.4 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.4.1 Polyethylene bagged packages. Polyethylene bagged packages shall have the required identification information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification. The bag or label shall only indicate the following information:

STOCK NUMBER
NOMENCLATURE
SIZE
QUANTITY

6. NOTES

6.1 Intended use. The liner is intended to be used in conjunction with the Trouser, Firemen's, Aluminized, Proximity and Trouser, Firemen's, Structural/ Aluminized Proximity.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Class, sizes, and lengths required (see 1.2).
c. When first article is required (see 3.2), the item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, quantity, and testing and approval of the first article.
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
e. When palletization is required (see 5.3).

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6.3 Samples and patterns. For access to samples and patterns, address the procuring activity issuing the invitation for bids.

6.4 Subject term (key word) listing.
Batting, aramid
Batting, novoloid
Liner, trousers
Liner, firemen's, proximity
Liner, firemen's, proximity/structural

Custodian:
Navy - NU
Air Force - 99

Review Activity:
DLA - CT
Navy - MC
Air Force - 50, 82

User Activity:
Navy - SH, CG
Air Force - 45

Preparing Activity:
Navy - NU

Project No. 8415-0626

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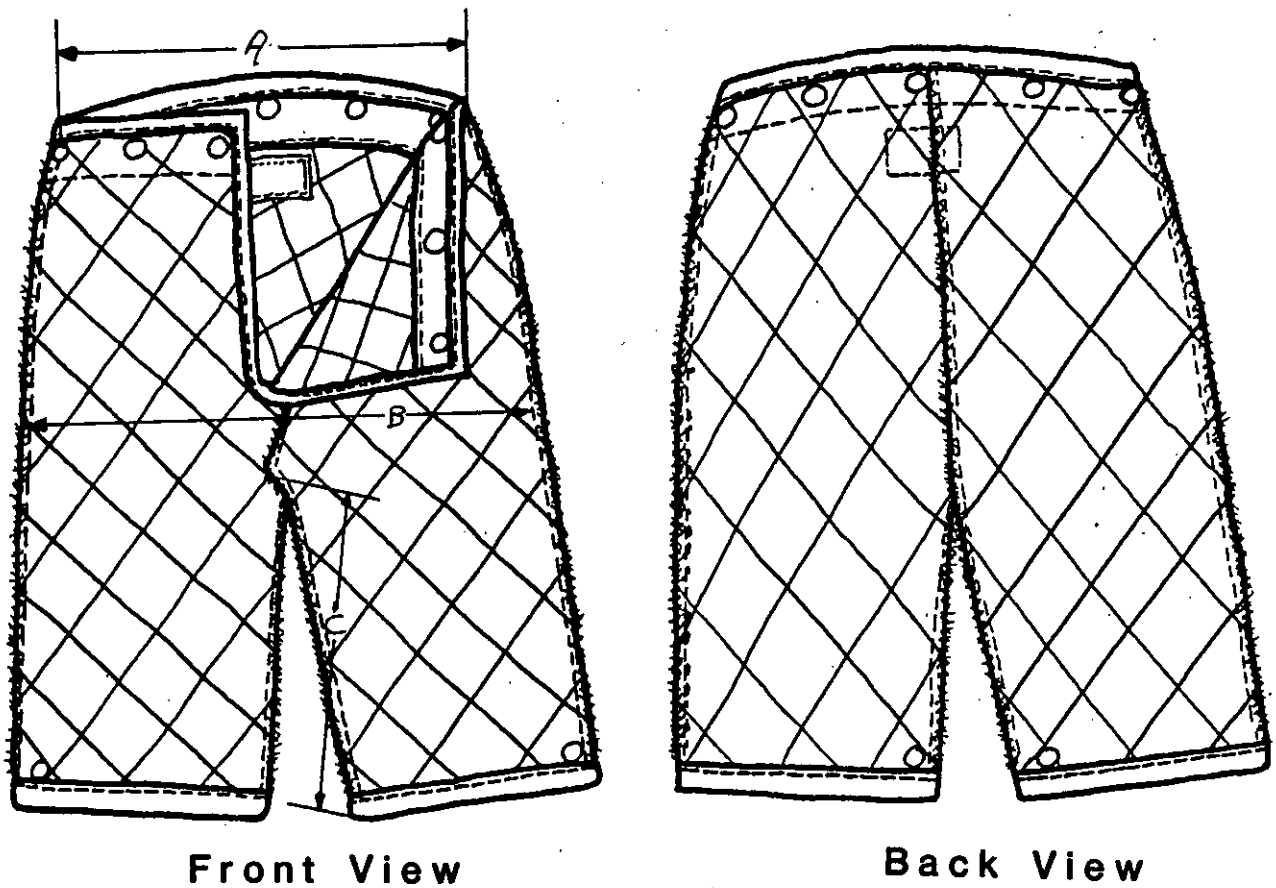


FIGURE 1
LINER, FIREMAN'S TROUSER, QUILTED

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-24942	LINER, FIREMEN'S TROUSER, QUILTED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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